

Latest DAQ equipment you might want to get your hands on

Global players, such as HBM, Spectrum Technologies and ACR Systems, have penetrated the Indian market, providing broader range of advanced and economical instruments, through various Indian representatives. Now you can even turn your notebook computer into a time-frequency spectrum analyser. The 24-bit dynamic signal-acquisition module, USB-2405, from Adlink Technology, along with AnCAD Technology's Visual Signal DAQ Express time-frequency analysis (TFA) software, enables machine diagnostics and failure prevention, research and portable field measurements on your notebook.

Another interesting product is Dynamic DAQ, developed by National Technical Systems, which allows collection and portrayal of data in real time from aerospace applications. It can be scaled to fit requirements of aerospace test stands and complex data sets, while still being able to measure hundreds of temperature and pressure readings.

Then there is a multi-sensor measurement module, called DT9829, which can accept different kinds of signals: current, voltage, thermocouple, RTD, thermistor or strain gauge. This module by Data Translation supports two to eight channels, where each channel can handle different sensors.

Measurement Computing introduced a high-speed DAQ board, which connects through USB and provides two 12-bit ADCs sampling 20G samples/s per channel simultaneously.

Another device from the same company, the BTH-1208LS, allows Bluetooth communication in a multifunction DAQ device. It offers short-range wireless DAQ to a compatible host device (Android-based tablet, phone or mini-PC or Windows-based PC). It can also operate as a physically-connected USB device.

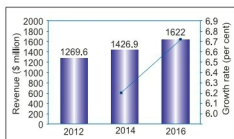


Fig. 1: Global revenue forecast (DAS)

which allows user-defined measurements, analysis and logging. A successful, software-defined DAQ system comprises modular, reconfigurable hardware platforms based on open industry standards, and a systems design software that can be used to easily define the functionality of the system.

Navjodh Dhillon, application engineer, Electronic Measurement Group, Agilent Technologies says, "Modular systems give you a facility of changing the module as and when required while making minimum changes to the overall DAQ system. They also allow you to put different kinds of modules in a small space, to make one integrated, unified system for DAQ purpose.

Today, DAQ is not just about only taking one type of measurement, it's taking 'n' types of different measurements. For this you need an instru-

ment which can have multiple kinds of modules that can be added to the instrument and still be able to take 'n' types of different measurements in a unified, integrated solution. This is how modular solutions help you out."

Improved sensor accuracy using digital multimeter technology

Many DAQ systems now include digital isolators to protect the device, computer and operator from potentially hazardous electrical signals while improving measurement accuracy. This is done by breaking ground loops and

rejecting error-causing common-mode voltages and other signal conditioning features such as amplifiers, filters and sensor-specific circuitry on new devices to further improve accuracy and simplify connectivity to common sensors.

Today, the major requirement for any DAQ system is to be as accurate as possible while continuing to be available at a minimal cost. For this, digital multimeter (DMM) based systems have come up. Many systems use DMM as the measurement instrument, which ensures minimum cost, and the measurements are as accurate as a standalone DMM would have.

Integrated signal conditioning

While the fundamental architecture of hardware has remained the same, the technologies used in various integrated components have evolved consistently. High-performance DAQ hardware with integrated signal conditioning have surfaced as next-generation solutions.

These solutions integrate DAQ and signal conditioning on a single module to provide high-performance, high-channel count and reliable sensor measurements. Most of such solutions are built around the industry-standard PXI platform, taking the advantage of the superior synchronisation and the high-throughput PCI express bus.

A modular, re-configurable, platform-based approach to building DAQ systems has some very clear advantages for the vendors

Productivity. With software-defined, modular instrumentation, the test engineer has the freedom to quickly mix and match various COTS hardware and software to customise a generic DAQ system to cater to multiple DAQ applications. This drastically increases the productivity of the test engineer, which is very much required in today's scenario with more and more products getting released every day.

Cost optimisation. Operational test costs are a combination of development costs, deployment costs and operational costs. A platform and ecosystem-based approach, and hardware and software reuse for multiple tests, reduce development costs because of less hardware requirement, reuse of hardware for different test setups, reduce operational costs because of lesser space requirements as modular instruments have a compact form factor and reduce deployment cost by driving down the man hours required for test and reducing the need of specialised training for different tools.

—Chinmay Misra, technical marketing engineer, National Instruments